Alibaba Cloud
AI Hackathon
Pakistan 2026

Project Re-Evaluation Proposal

Grade 2A to Grade 1 Re-Evaluation

Scam Detection ML System

Multilingual SMS Fraud Classification

Submitted: 28 August 2026

Build Phase Deadline: 4 September 2026

Alibaba Cloud AI Hackathon Pakistan 2026 -- Re-Evaluation Proposal

1. Problem Statement

Pakistan loses billions of rupees annually to SMS and messaging-based fraud. The Federal Investigation Agency (FIA) reports that scam messages targeting mobile wallets (JazzCash, EasyPaisa), banking customers, and government welfare beneficiaries (BISP, Ehsaas) are the most prevalent cybercrime vector in the country.

The problem is uniquely difficult in Pakistan because:
- **Linguistic diversity**: Scam messages arrive in English, Roman Urdu (informal Urdu written in Latin script with no standard spelling), Urdu script, and mixed-language code-switching -- often within the same message.
- **Evolving tactics**: Scammers rapidly adapt, using brand impersonation (HBL, JazzCash, NADRA, FBR), urgency/threat psychology, and culturally-specific hooks (BISP stipends, SIM registration, overseas job offers).
- **Scale**: With 190+ million mobile subscribers, manual detection is impossible. Automated, real-time classification is essential.

Our project: A multilingual ML system that classifies text messages as Scam or Safe across all 4 Pakistani language contexts, with production-grade accuracy and a clear path to real-time deployment.

2. Proposed Solution

We have built and validated a complete scam-detection ML pipeline -- not a prototype, but a working system with verified performance across 4 independent test sets.

Solution Architecture

```
User Message (SMS / WhatsApp / App)
  |
  v
+---------------------------------------------+
|  REST API (FastAPI) -- planned by Sep 4      |
|  +- Single endpoint: POST /classify         |
|     Returns: {label, probability, category} |
+------------------+--------------------------+
                   |
                   v
+---------------------------------------------+
|  ImprovedScamTextNormalizer                 |
|  - Unicode NFKC normalization               |
|  - Roman Urdu spelling normalization (15+   |
|    rules: btayen->batayein, krin->karein...)    |
|  - USSD code normalization (*786# -> token)  |
|  - Phone/URL/currency pattern preservation   |
|  - Character collapse (freeeee -> free)      |
+------------------+--------------------------+
                   |
                   v
+---------------------------------------------+
|  Feature Extraction (dual-stream)           |
|                                             |
|  Stream A: Word TF-IDF                      |
|    n-grams (1,2), max 30,000 features       |
|    sublinear TF, unicode accent stripping    |
|                                             |
|  Stream B: Character TF-IDF                 |
|    char_wb n-grams (3,5), max 30,000 feats  |
|    captures sub-word patterns across langs   |
|                                             |
|  [FeatureUnion -> sparse concatenation]      |
+------------------+--------------------------+
                   |
                   v
+---------------------------------------------+
|  Classifier: LinearSVC (C=5.0)             |
|  - class_weight="balanced"                  |
|  - Platt scaling for calibrated probability |
|  - Optimized threshold: 0.49                |
|                                             |
|  -> Scam Probability >= 0.49 -> "Scam"        |
|  -> Scam Probability < 0.49 -> "Safe"        |
+---------------------------------------------+
```

Key Differentiators

1. Pakistan-specific domain knowledge embedded in preprocessing: 150+ multilingual scam-signal keywords across 6 categories (urgency, financial, credential, prize, threat, OTP), recognition of 60+ Pakistani brands (telecoms, banks, government services, health institutions), and "do-not-share" safe-signal patterns.

2. 39 engineered features beyond TF-IDF: message statistics, digit ratios, keyword scores, suspicious domain detection, legitimate brand recognition, personal tone scoring, service notification detection, receipt pattern matching.

3. Hard-negative mining: 82 adversarial safe messages specifically designed to mimic common false-positive triggers (bank deduction notifications, "Congratulations selected" for legitimate programs, security alerts, financial statements).

4. **Group-aware leakage-safe splitting**: 4-word prefix grouping prevents template-based data leakage between train/val/test splits.

3. Detailed Project Description

Current Implementation Status: PRODUCTION-READY CORE

The entire ML pipeline is implemented, tested, and validated. This is not a concept -- it is working code.

Module Breakdown (12 modules, 2,500+ lines)

Module	File	Lines	Function
Configuration	`src/config.py`	39	Centralized constants, paths,
Data Audit	`src/data_analysis.py`	258	Dataset loading, quality audit
Preprocessing	`src/preprocessing.py`	334	Two-tier normalizer: base Scam
Feature Engineering	`src/features.py`	283	39-engineered numeric features
Training	`src/train.py`	460	Stratified splitting, 5-fold C
Evaluation	`src/evaluate.py`	523	Multi-model comparison, thresh
Prediction API	`src/predict.py`	171	Single-message classification
V3 Pipeline	`run_retrain_v3_pipeline.py`	1,205	Production pipeline: 6 model c
Orchestration	`run_pipeline.py`	250	End-to-end reproducibility orc
External Validators	`tests/`	8 files	4 independent validation suite

Model Selection Process

6 candidate architectures were evaluated with 5-fold stratified cross-validation:

Model	Architecture	CV F1	Test Accuracy	Test F1	ROC-AUC
A_combined_C2	Word+Char TF-IDF, Lin	0.9473	96.79%	0.9706	0.9963
B_combined_C5	Word+Char TF-IDF, Lin	0.9474	97.33%	0.9756	0.9960
C_wide_char	Word+Char(3,6) TF-IDF	0.9441	96.79%	0.9709	0.9962
D_combined_lr	Word+Char TF-IDF, Log	0.9539	94.12%	0.9447	0.9954
E_calibrated_svm	Calibrated LinearSVC (i	0.9515	94.65%	0.9533	0.9954
F_trigram_combined	Word(1,3)+Char(3,6) TF	0.9452	96.79%	0.9703	0.9966

Selected: B_combined_C5 -- best composite score (60% F2 + 40% Accuracy) with threshold 0.49.

Validated Performance Across 4 Independent Test Sets

Test Set	Messages	Accuracy	F1	Recall	Precision	ROC-AUC	FP	FN
Internal Holdou	187	97.33%	0.9756	97.09%	98.02%	0.9960	2	3
All-4-Language	318	97.48%	0.9798	96.52%	99.49%	0.9968	1	7
Truly Blind (fres	50	98.00%	--	96.00%	100%	--	0	1
Real-World Edg	43	93.02%	--	87.50%	100%	--	0	3

Per-Language Performance (318-message external test)

Language	Samples	Accuracy	Precision	Recall	F1
English	118	99.15%	100%	98.00%	0.9899
Mixed	68	98.53%	97.96%	100%	0.9897
Urdu	66	96.97%	100%	95.74%	0.9783
Roman Urdu	67	94.03%	100%	91.67%	0.9565

Overfitting Verification

- Cross-validation accuracy: 94.21%
- Test accuracy: 97.33%
- CV-test gap: **-1.82%** (excellent generalization, < 3% threshold)
- No significant overfitting detected

4. Technical Approach and Technologies

Technology Stack

Layer	Technology	Rationale
Language	Python 3.10+	ML ecosystem maturity
ML Framework	scikit-learn 1.3+	Reproducibility, interpretabil
Feature Extraction	TF-IDF (word + character n-gra	Language-agnostic, fast infere
Classifier	LinearSVC (C=5.0)	Optimal accuracy-speed tradeof
Calibration	Platt scaling (sigmoid on deci	Calibrated confidence scores f
Data Processing	pandas, numpy, scipy	Efficient data handling and sp
Persistence	joblib	Fast model serialization/deser
API (planned)	FastAPI + Uvicorn	Async inference, production de
Frontend (planned)	React + Tailwind CSS	User-facing demo for hackathon
Deployment (planned)	Docker + Alibaba Cloud ECS	Scalable, cloud-native

Why TF-IDF + SVM Instead of Transformers?

This is a deliberate engineering decision, not a limitation:

1. **Latency**: TF-IDF+SVM inference is <10ms per message vs. 50-200ms for transformer models. For real-time SMS filtering at telecom scale (millions of messages/day), this matters.
2. **Data efficiency**: With 1,123 training messages, fine-tuning mBERT or XLM-R would overfit. TF-IDF+SVM generalizes better at this data scale (verified: CV-test gap < 3%).
3. **Interpretability**: Feature importance can be traced back to specific n-grams and engineered features, enabling forensic analysis of false positives/negatives.
4. **Resource efficiency**: No GPU required. Runs on any cloud VM, including Alibaba Cloud's basic ECS instances.
5. **Multilingual handling**: Character n-grams (3,5) naturally capture sub-word patterns across all 4 languages including Urdu script, without requiring language-specific tokenizers.

Planned by Sep 4: Add a transformer comparison arm (XLM-RoBERTa-base fine-tuned on our dataset) to benchmark against our TF-IDF+SVM baseline, demonstrating methodological rigor.

Data Pipeline Integrity

```
Raw Dataset (868 messages)
  |
  +- Data Quality Audit
  |    - Missing value detection
  |    - Exact duplicate removal
  |    - Near-duplicate detection (SequenceMatcher, 90% threshold)
  |    - Conflicting label detection (same text, different label)
  |    - Control character detection
  |
  +- V3 Augmentation (+255 messages)
  |    - 82 hard-negative safe messages (adversarial)
  |    - 173 scam messages across 10 new categories
  |    - Roman Urdu spelling variation injection
  |
  +- Group-Aware Splitting
  |    - 4-word prefix grouping prevents template leakage
  |    - Stratified: Train 795 / Val 141 / Test 187
  |    - Verified: 0 group overlap between splits
  |
  +- 5-Fold Stratified Cross-Validation
       - Fresh model initialization per fold
       - Train + test metrics tracked for overfitting detection
```

5. Delivery Plan to September 4 (Build Phase Close)

What is Already Done (Day 0 -- Today)

[x] Complete ML pipeline (12 modules, 2,500+ lines)
[x] 6 model architectures evaluated and compared
[x] Production model selected and saved (B_combined_C5)
[x] 4 independent validation suites passing at 93-98% accuracy
[x] Multilingual support (English, Roman Urdu, Urdu, Mixed)
[x] Comprehensive error analysis and overfitting verification

[x] Prediction interface (CLI + programmatic API)
[x] Reproducible pipeline (single command: `python run_retrain_v3_pipeline.py`)

Week 1 (Aug 29 -- Sep 4): Remaining Deliverables

Task	Deadline	Description
REST API	Aug 30	FastAPI wrapper around `predic
Web Dashboard	Sep 1	React frontend: paste a messag
Transformer Comparison	Sep 2	Fine-tune XLM-RoBERTa-base on
SMS Integration Demo	Sep 3	Twilio/Alibaba Cloud SMS integ
Alibaba Cloud Deployment	Sep 3	Containerize (Docker), deploy
Presentation Prep	Sep 4	Slide deck, live demo script,

Post-Hackathon Roadmap (Regional Round)

Feature	Timeline
Audio/call scam detection (Whi	Q4 2026
Real-time SMS feed integration	Q4 2026
Continuous learning pipeline (	Q1 2027
Multi-language model distillat	Q1 2027

6. Why This Project Deserves Grade 1

1. It works, today. This is not a slide deck or a Jupyter notebook. It is a 2,500+ line production-grade ML system with saved model artifacts, reproducible pipelines, and 4 independent validation suites proving generalization.

2. 97.48% accuracy on 318 unseen messages across 4 languages -- including Roman Urdu, one of the hardest languages for NLP due to extreme spelling variation and code-switching.

3. Scientific rigor. 6 candidate models compared. 5-fold CV. Group-aware leakage-safe splitting. Near-duplicate detection. F2-optimized thresholds. Overfitting verification. Error analysis. These are the practices of a mature ML project.

4. Real social impact. Pakistan's telecom subscribers (190M+) are the target audience. Every correctly classified scam message potentially saves a family from losing their savings.

5. Clear deployment path. The architecture is designed for cloud deployment (no GPU needed, <10ms inference, REST API ready). The Sep 4 deliverables include a live demo on Alibaba Cloud.

6. Honest assessment of limitations. We document exactly where the model fails (Roman Urdu edge cases at P=0.42-0.48 threshold boundary), why we made our engineering choices (TF-IDF vs. transformers), and what comes next.

This project was built entirely during the Alibaba Cloud AI Hackathon Pakistan 2026 build phase. All code, data, models, and reports are reproducible from the repository.